G.O.S. sensor node rev 3.1

18650 cell + USB-C charging, nRF52840 module (Thread / BLE), SHT40 and TSL2591 over I2C.
LED drivers live on a separate, separately powered board; J4 gives it the two PWM signals.
The 18650 sits in the enclosure and plugs into J5. LoRa and GNSS are not fitted.

Sheets share nets through global labels and power symbols; the sheet symbols below carry no pins.
Pin map, power tree and datasheet sources: HARDWARE.md

Power

File: power.kicad_sch

MCU

File: MCU.kicad_sch

Sensors

File: sensors.kicad_sch

Connectors

File: connectors.kicad_sch

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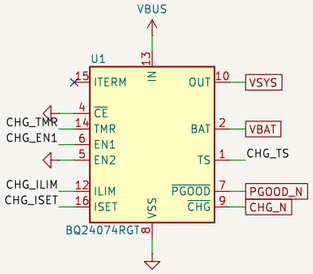
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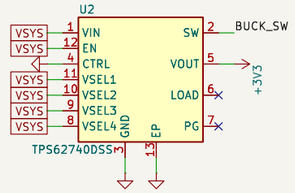
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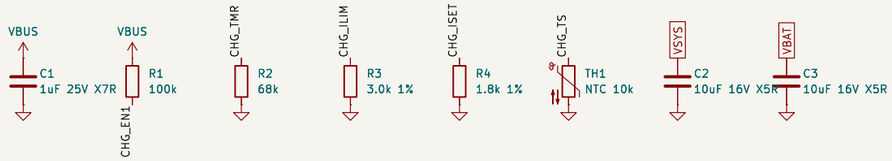
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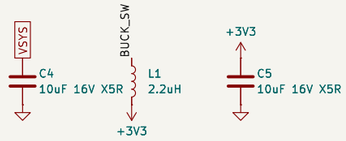
Size: A4	Date:	**Rev: 3.1**
KiCad E.D.A. 9.0.7		Id: 1/5

USB-C 5 V -> BQ24074 (charger + power path, VSYS = 4.4 V on USB, VBAT on battery) -> TPS62740 (3.3 V, all four VSEL high)
USB500 mode: EN2 = GND, EN1 high only while VBUS is present (no battery drain through the EN pull-down).
Charge 494 mA (ISET 1.8k), safety timer 9.1 h (TMR 68k), termination 10 % (ITERM open).
TH1 sits beside J5: it senses board temperature next to the cell, not the cell itself. Without a thermistor fit a fixed 10k.

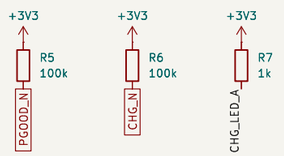


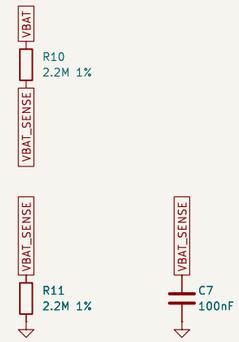




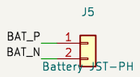


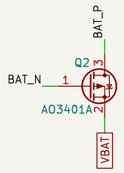


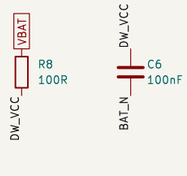


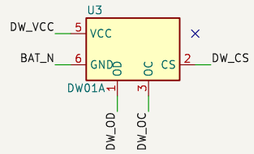


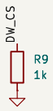
Cell protection. BAT_N is the cell negative, NOT system GND: the FS8205A sits between them.
DW01A trips: overcharge 4.30 V, overdischarge 2.40 V (release 3.0 V), overcurrent 150 mV across both FETs (about 2.5 A).
The 18650 sits in the enclosure cradle and plugs into J5 (pin 1 = positive). Q2 blocks a reversed lead.
After connecting a cell, plug USB in once to wake the DW01A.

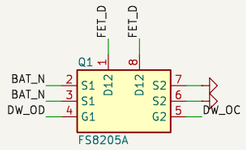












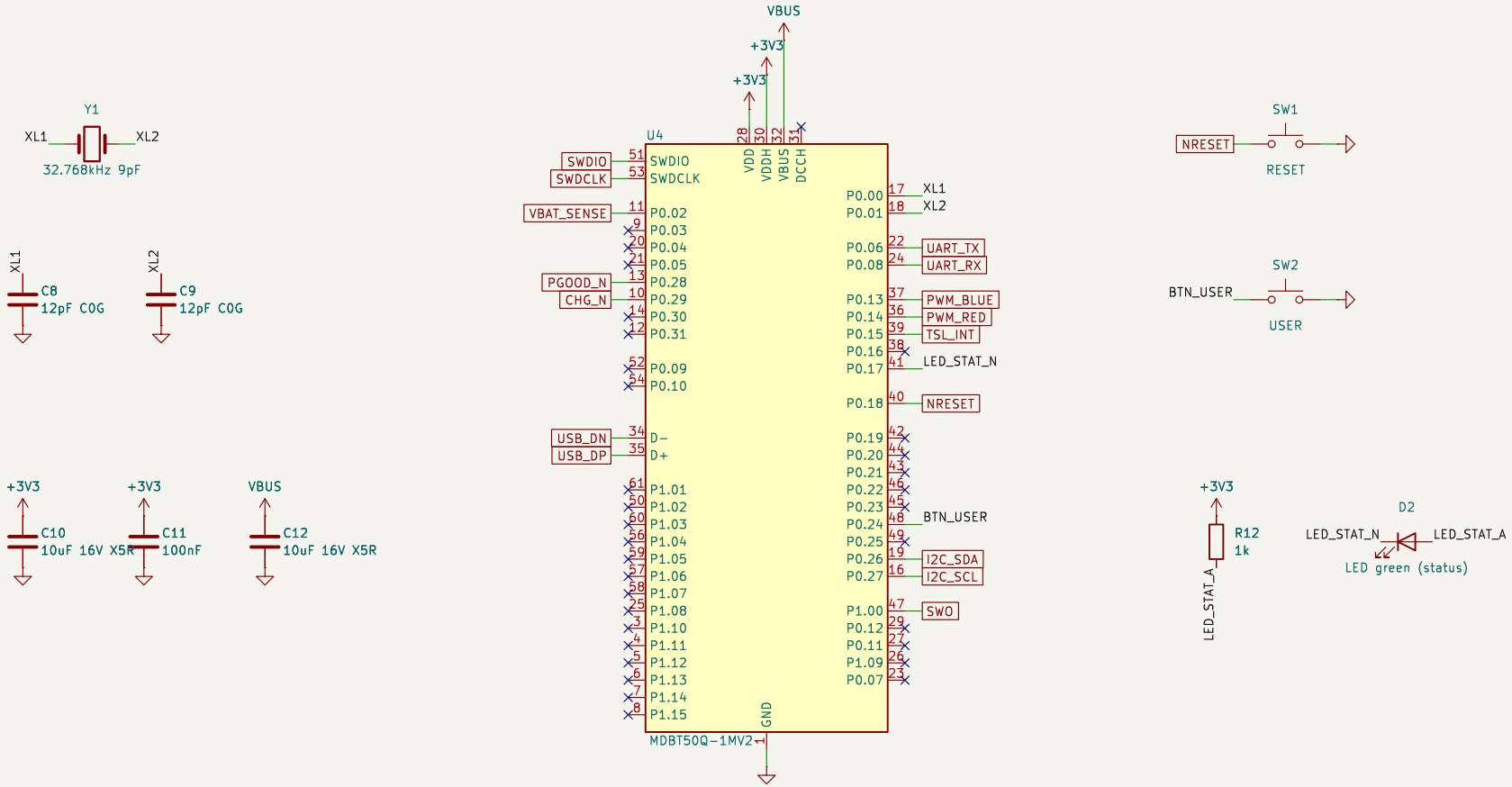
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Sheet: /Power/
File: power.kicad_sch

Title: Power: USB charger, 3.3 V buck, cell protection

Size: A3	Date:	**Rev: 3.1**
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VDD and VDDH tied to 3.3 V, DCCH open: Raytac reference circuit 8.3 (REG0 off, supply below 3.6 V).
Module pins 3–14 and 50–61 are low-frequency only (below 10 kHz): no SPI, I2C, UART or PWM there.
Y1: 32.768 kHz, 9 pF load, with 12 pF caps as in the Raytac reference.
Keep copper out of the module antenna area (Raytac spec section 2.3).

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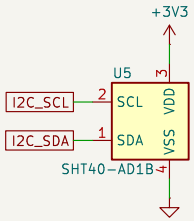
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Sheet: /MCU/
File: MCU.kicad_sch

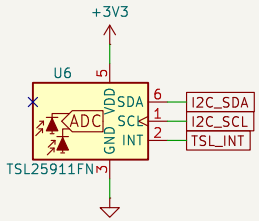
Title: MCU: Raytac MDBT50Q-1MV2 (nRF52840)

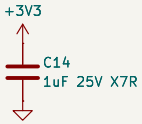
Size: A4	Date:	**Rev: 3.1**
KiCad E.D.A. 9.0.7		Id: 3/5

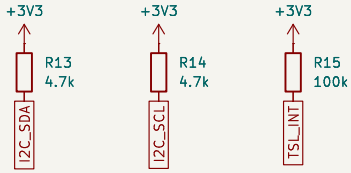
I2C addresses: SHT40 0x44, TSL2591 0x29. Place U5 away from the charger and buck (self-heating).
U6 needs a clear view of the light: no part taller than it nearby. U6 needs at least 2.7 V and is rated -30 to 70 C.
No copper under U5 other than its four pads (Sensirion land pattern); its centre pad stays unsoldered.











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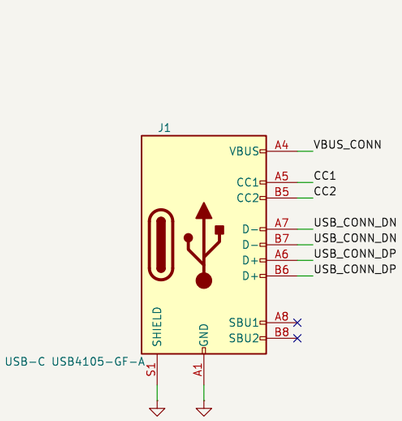
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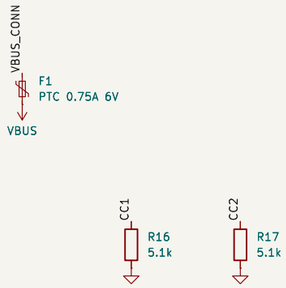
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File: sensors.kicad_sch

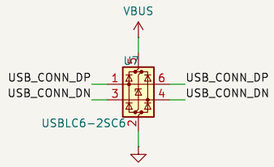
Title: Sensors: SHT40, TSL2591

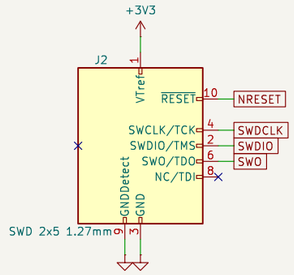
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KiCad E.D.A. 9.0.7		Id: 4/5

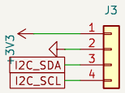
J1: USB 2.0 device. 5.1k on each CC pin asks the host for 5 V.
J4 carries only the two PWM signals to the separate LED driver board, which has its own 5 to 12 V supply (AL8860 needs at least 4.5 V; do not power it from this board). 100k pull-downs hold the signals low during reset.
J2: standard 10-pin 1.27 mm Cortex debug header.

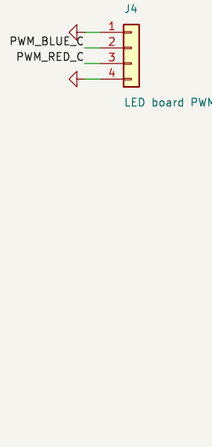




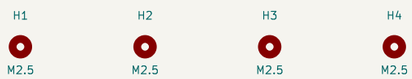












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Sheet: /Connectors/
File: connectors.kicad_sch

Title: Connectors: USB-C, SWD, I2C, LED board, test points

Size: A3	Date:	**Rev: 3.1**
KiCad E.D.A. 9.0.7		Id: 5/5